

CLINICAL DENTAL PRACTICE GUIDELINE: POST-EXTRACTION SURGICAL PROTOCOL, BLEEDING CONTROL, AND DRY SOCKET MANAGEMENT

Oral and Maxillofacial Surgery Clinical Protocols

Guideline Code: DENT-SURG-2024-V1

1. ROUTINE POST-EXTRACTION CARE & HEMOSTASIS

- Primary Hemostasis:

* Place sterile folded gauze directly over extraction socket. Instruct patient to apply firm, steady biting pressure for 30 to 45 minutes continuously.

* Avoid spitting, rinsing, drinking through straws, smoking, or vigorous exercise for 24 hours to prevent dislodging the initial blood clot.

* Ice packs applied externally (20 minutes on, 20 minutes off) for the first 24 hours to minimize postoperative edema.

- Secondary Hemostasis for Persistent Oozing:

* If mild bleeding persists after 45 minutes, replace with a damp black tea bag wrapped in gauze and bite firmly for 30 minutes. Tannic acid in tea promotes local platelet aggregation and vasoconstriction.

* For persistent socket bleeding: Inspect socket under good illumination. Clean excess liver clot with suction. Place topical hemostatic agents:

- Oxidized regenerated cellulose (Surgicel) or absorbable gelatin sponge (Gelfoam) into socket.

- Place figure-of-eight 3-0 or 4-0 resorbable suture (Vicryl or Chromic Gut) across the socket margins.

- Local injection of 2% lidocaine with 1:100,000 epinephrine directly into surrounding gingival papillae for vasoconstriction.

* Topical Tranexamic Acid (TXA): 5% rinse (crush 500 mg tablet in 10 mL water) or soak gauze in TXA solution and hold under pressure for 20 minutes.

2. MANAGEMENT OF PATIENTS ON ANTICOAGULANTS OR ANTIPLATELET THERAPY

- Warfarin (Coumadin): Check INR within 24 hours before procedure.

* Therapeutic INR range between 2.0 and 3.0: Routine extractions can proceed safely without discontinuing Warfarin. Do NOT stop Warfarin due to rebound thromboembolism risk.

* Use local hemostatic measures (Gelfoam/Surgicel, TXA mouthwash, primary closure with sutures).

- Direct Oral Anticoagulants (DOACs - Apixaban, Rivaroxaban, Dabigatran):

* For standard single or simple extractions, DOAC discontinuation is generally unnecessary.

* Schedule surgery 12 to 24 hours after last dose; next dose taken after definitive hemostasis achieved.

- Antiplatelet Agents (Aspirin 81 mg, Clopidogrel 75 mg): Maintain therapy. Rely on local hemostatic agents.

3. DRY SOCKET (ALVEOLAR OSTEITIS)

- Definition: Premature breakdown and fibrinolysis of blood clot within extraction socket, exposing alveolar bone and nerve endings.

- Incidence: 1-3% of routine extractions, up to 25-30% of impacted mandibular third molars.

- Risk Factors: Tobacco smoking, oral contraceptives, difficult traumatic extraction, poor oral hygiene, female gender.
- Clinical Presentation:
 - * Severe, throbbing, radiating pain developing 2 to 4 days post-operatively.
 - * Pain often radiates to ear, temple, or eye.
 - * Characteristic foul odor (halitosis), metallic bad taste, and visibly empty socket denuded of clot with exposed grayish bone.
- Treatment Protocol:
 1. Gently irrigate socket with warm sterile saline or 0.12% Chlorhexidine to flush necrotic food debris. Do NOT aggressively curette the bone, as this causes further bone trauma and delays healing.
 2. Place a small sedating dressing: Zinc Oxide Eugenol paste on iodoform gauze or Alveogyl (eugenol-impregnated fiber dressing).
 3. Change dressing every 24 to 48 hours until pain resolves (usually 3 to 6 days).

4. Prescribe NSAIDs (Ibuprofen 600-800 mg TID with food) for analgesia.

5. Note: Antibiotics are NOT routinely indicated for dry socket unless there is spreading cellulitis, fever, or trismus.